

Experiment 8: Steering Angle Variation in MIMO Beamforming Using MATLAB

Objectives, MATLAB Programs and Output Images

Objective 1

To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.

MATLAB Program

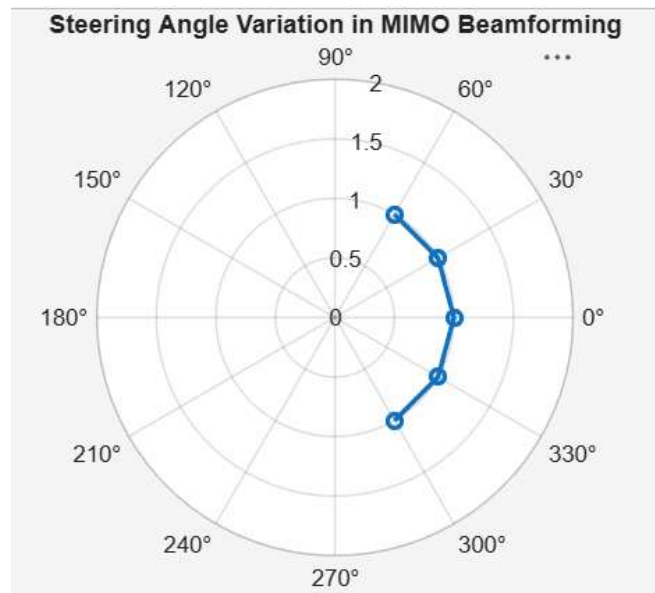
```
clc;
clear;
close all;

angle = [-60 -30 0 30 60];          % Steering Angles (degrees)
gain = ones(size(angle));           % Normalized Gain

figure
polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```

Output Image – Objective 1



Objective 2

To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.

MATLAB Program

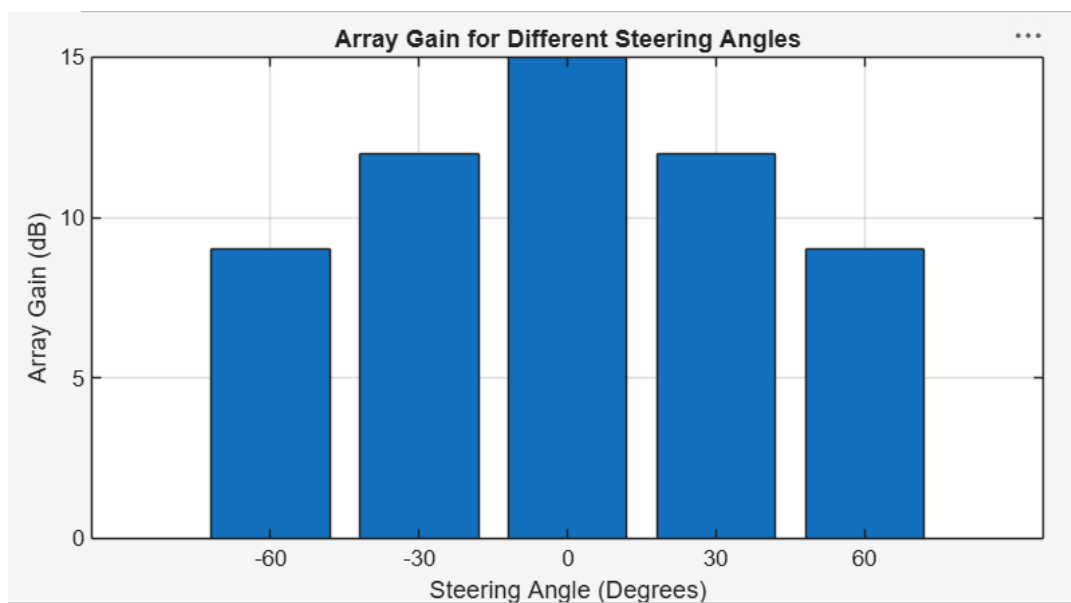
```
clc;
clear;
close all;

angle = [-60 -30 0 30 60];           % Steering Angles (degrees)
gain = [9 12 15 12 9];               % Array Gain (dB)

figure
bar(angle, gain)

grid on
xlabel('Steering Angle (Degrees)')
ylabel('Array Gain (dB)')
title('Array Gain for Different Steering Angles')
```

Output Image – Objective 2



Objective 3

To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

MATLAB Program

```
clc;
clear;
close all;

desired = [-60 -30 0 30 60];           % Desired Steering Angle (degrees)
actual = [-58 -32 1 29 62];           % Actual Beam Direction (degrees)

error = abs(desired - actual);         % Beam Direction Error

figure
plot(desired, error, 'o-', 'LineWidth', 2)

grid on
xlabel('Desired Steering Angle (Degrees)')
ylabel('Beam Direction Error (Degrees)')
title('Beam Direction Error for Different Steering Angles')
```

Output Image – Objective 3

